

SensoryWardrobe — Requirements Document

Prepared by: Bruce Schulz, Nova Denton-Parry, Zach Kahler, and Jeremy Kirkpatrick

Program: Computer Programming, Southeast Technical College

Date: June 2026

1. Problem and Solution

a. Problem Description

Every morning, sensory-sensitive individuals — including those with Sensory Processing Disorder (SPD), autism, and ADHD — face a uniquely exhausting challenge: choosing what to wear. Unlike neurotypical individuals who may simply glance at the weather forecast and grab an outfit, sensory-sensitive users must weigh fabric texture, seam placement, pressure points, layering weight, and how all of these interact with the day's temperature and humidity. A single poor choice can lead to sensory overload that derails productivity, mood, and daily functioning for hours.

Currently, no automated information system exists that combines real-time weather data with a user's personal sensory-comfort history to recommend clothing. Standard weather apps report temperature and precipitation but have no awareness of how a fleece jacket feels against sensitive skin on a humid 55°F day versus a dry one. Manual journals are unsearchable and lack weather correlation. The result is chronic decision fatigue and frequent discomfort for millions of people.

The opportunity is clear: an automated, data-driven system that logs outfit choices alongside weather conditions, captures comfort ratings with sensory-specific tags, and — over time — surfaces historically successful outfits for similar weather days. This transforms a daily struggle into a guided, confidence-building routine.

b. Organization Context

This project originates from the Computer Programming program at Southeast Technical College. The development team consists of Bruce Schulz, Nova Denton-Parry, Zach Kahler, and Jeremy Kirkpatrick. The target market for the end product is the broader assistive-technology and wellness-app space serving neurodivergent communities, occupational therapy clinics, and caregivers of sensory-sensitive children. The app fills a gap where healthcare meets daily-living technology — an underserved intersection with growing awareness and demand.

2. Chosen Solution

SensoryWardrobe is a cross-platform mobile application built with Flutter (Dart) that functions as a weather-aware clothing tracker for sensory-sensitive users. The system provides:

- **Automatic Weather Fetching** — Integrates with the OpenWeatherMap API to pull the current day's forecast (temperature, humidity, wind speed, conditions) based on the user's location.
- **Outfit Logging** — Users select clothing items from a personal wardrobe catalog, tagging each item with sensory attributes (texture, pressure level, seam type, weight).
- **Comfort Rating** — After wearing an outfit, users rate comfort on a 1–5 star scale and can annotate specific sensory pain points.
- **Smart Suggestions** — The app queries a local SQLite database for previously logged outfits that received high comfort ratings on days with similar weather profiles, presenting them as morning recommendations.
- **Personalized Growth** — Over weeks and months the system builds a personalized clothing-decision guide unique to each user's sensory profile.

The system stores all data locally on-device for privacy, with optional encrypted cloud backup. It is designed with accessibility-first principles: high-contrast modes, large tap targets, minimal cognitive load per screen, and screen-reader compatibility.

3. Alternative Solutions

Alternative 1: Manual Paper Journal

A user could maintain a handwritten notebook logging each day's outfit, the weather, and a comfort note. This approach is zero-cost and requires no technology. However, it is extremely difficult to search historical entries, impossible to automatically correlate with weather data, and prone to being lost or abandoned. It offers no recommendation capability and requires the user to remember to check previous entries — adding cognitive load rather than reducing it.

Alternative 2: Spreadsheet (Google Sheets / Excel)

A semi-automated approach using a spreadsheet with columns for date, weather (manually entered or pulled via a formula), clothing items, sensory tags, and comfort score. While flexible and sortable, this approach requires significant manual data entry each day, provides no push notifications or morning suggestions, and is cumbersome on mobile devices. The friction of daily use typically leads to abandonment within weeks. Filtering for "similar weather days" requires manual formula construction beyond most users' comfort level.

Alternative 3: Generic Outfit/Style Apps (e.g., Stylebook, Cladwell)

Existing wardrobe apps allow users to photograph clothing and plan outfits. Some integrate weather forecasts. However, none include sensory-specific tagging (texture, pressure, seam irritation), none capture comfort ratings tied to weather conditions, and none provide recommendations based on historical sensory-comfort data. They solve a fashion-coordination problem, not a sensory-accessibility problem. Their recommendation engines optimize for style variety, not comfort consistency.

4. Why This Solution Is Unique

SensoryWardrobe is unique because it is the only system that:

1. **Links weather conditions to personal sensory comfort** — not just "it's cold, wear a coat" but "on humid 50°F days, your soft cotton hoodie rated 5 stars while your wool sweater rated 2 stars."
2. **Uses sensory-specific tags as first-class data** — texture, pressure, seam placement, weight, and tightness are core attributes in every clothing record, not optional notes.
3. **Learns and improves over time** — the recommendation algorithm weights recent ratings more heavily, adapting as the user's sensory preferences evolve.
4. **Addresses an underserved population** — neurodivergent individuals and their caregivers have no existing digital tool purpose-built for this daily-living need.
5. **Minimizes cognitive load by design** — every screen is built for users already experiencing decision fatigue; logging takes under 60 seconds, and suggestions are surfaced proactively via push notification.
6. **Combines automation with personalization** — weather is fetched automatically (no manual entry), but recommendations are deeply personal (based on the individual's own history, not generic advice).

5. Scope of the Solution

The scope of SensoryWardrobe for this initial release includes the following functional areas:

Functional Area	Included Features
User Account	Registration, login, password reset, profile creation with sensory preferences
Wardrobe Catalog	Add/edit/delete clothing items with photo, category, warmth level, and sensory tags
Weather Integration	Automatic daily forecast retrieval via OpenWeatherMap API; manual location override
Outfit Logging	Select items from wardrobe to compose a daily outfit; associate with current weather
Comfort Rating	1–5 star rating after wearing; optional sensory-pain-point annotations
Smart Suggestions	Morning recommendation of top-rated outfits from similar past weather days
History & Trends	Calendar view of past outfits with comfort scores; trend graphs over time
Accessibility	High-contrast mode, large tap targets, screen-reader labels, reduced-motion option
Data Storage	Local SQLite database; optional encrypted cloud backup and restore
Notifications	Configurable morning push notification with weather summary and outfit suggestion
Multi-Profile	Caregiver mode to manage separate profiles for children

Out of scope for this release: social/sharing features, therapist portal, integration with smart-home devices, e-commerce links, and wearable-device integration.

6. Product Backlog (User Stories)

The following 25 user stories form the initial product backlog. Each follows the format: *As a [role], I want [function], so that [value]*.

#	User Story
1	As a sensory-sensitive user, I want to create an account with my email and password so that my outfit data is saved and accessible only to me.
2	As a sensory-sensitive user, I want to see a "Forgot Password" button on the login screen so that if I cannot remember my password, I can reset it via email without losing my data.
3	As a sensory-sensitive user, I want to set my home location during onboarding by entering a city name or allowing GPS access so that the app can automatically fetch accurate local weather data each morning.
4	As a sensory-sensitive user, I want to add a clothing item to my wardrobe by entering its name, selecting a category (top, bottom, outerwear, undergarment, accessory), and uploading a photo so that I can select it when logging outfits.
5	As a sensory-sensitive user, I want to assign sensory tags (e.g., "soft cotton," "scratchy seams," "tight elastic," "heavy weight," "smooth lining") to each clothing item from a predefined list so that the app understands my sensory experience with that garment.
6	As a sensory-sensitive user, I want to assign a warmth level (1–5 scale) to each clothing item so that the app can factor thermal comfort into its weather-based recommendations.
7	As a sensory-sensitive user, I want the app to automatically fetch today's weather (temperature, humidity, wind speed, and conditions) when I open it each morning so that I do not have to enter weather data manually.
8	As a sensory-sensitive user, I want to compose a daily outfit by selecting multiple items from my wardrobe catalog on a single screen so that logging what I am wearing takes less than 60 seconds.
9	As a sensory-sensitive user, I want to rate my outfit's overall comfort on a 1–5 star scale at the end of the day so that the app can learn which combinations work for me.
10	As a sensory-sensitive user, I want to annotate my comfort rating with specific sensory pain points (e.g., "collar tag irritated neck," "waistband too tight after lunch") by selecting from common issues or typing a custom note so that the app captures detailed feedback beyond just a number.
11	As a sensory-sensitive user, I want to receive a configurable morning push notification showing today's weather and suggesting up to three outfits that scored 4+ stars on similar weather days so that I can start my day with confidence instead of anxiety.
12	As a sensory-sensitive user, I want to view a calendar screen showing each day's logged outfit and comfort score color-coded (green for 4–5, yellow for 3, red for 1–2) so that I can spot patterns over weeks and months.

#	User Story
13	As a sensory-sensitive user, I want to filter my outfit history by comfort rating (e.g., show only 4- and 5-star days) so that I can quickly find my best-performing combinations.
14	As a sensory-sensitive user, I want to filter my outfit history by weather range (e.g., 40–55°F and humid) so that I can see what worked on days similar to today.
15	As a sensory-sensitive user, I want to edit a clothing item's details (name, photo, tags, warmth level) so that my wardrobe catalog stays accurate when items change or I discover new sensory reactions.
16	As a sensory-sensitive user, I want to delete a clothing item from my wardrobe with a confirmation prompt so that I can remove donated or worn-out garments without accidental data loss.
17	As a sensory-sensitive user, I want to see a trend graph of my average weekly comfort score over the past 3 months so that I can visualize whether my clothing choices are improving over time.
18	As a sensory-sensitive user, I want to toggle a high-contrast dark mode in settings so that the app is comfortable to use when I am visually overstimulated.
19	As a sensory-sensitive user, I want all buttons and interactive elements to have a minimum tap target of 48×48 dp so that I can use the app without fine-motor frustration on days when my coordination is impaired.
20	As a sensory-sensitive user, I want to manually override my location for weather fetching by searching for a different city so that I can get accurate data when traveling or when GPS is unavailable.
21	As a caregiver, I want to create and switch between multiple user profiles within one account so that I can manage wardrobe and comfort data for my child separately from my own.
22	As a caregiver, I want to log an outfit and comfort rating on behalf of my child's profile so that I can build their comfort history even if they cannot use the app independently.
23	As a sensory-sensitive user, I want to set a preferred reminder time in settings so that the morning suggestion notification arrives when it is most useful in my routine.
24	As a sensory-sensitive user, I want to back up my data to encrypted cloud storage by tapping a "Backup Now" button in settings so that I do not lose my comfort history if I change or lose my phone.
25	As a sensory-sensitive user, I want to restore my data from a cloud backup onto a new device during setup so that I can continue using the app seamlessly after upgrading my phone.

7. Requirements Determination

a. Techniques Used for Gathering Requirements

The team employed three primary requirements-gathering techniques:

1. Interviews (Semi-Structured)

Semi-structured interviews were conducted with two individuals: both sensory-sensitive adults. The interviews were conducted virtually via a text chat, but the direct communication allowed more detailed answers than a survey. The questions (see Appendix A) were sent to the participants, and they were allowed to answer them at their own pace. Interviews allowed the team to explore lived experiences in depth, uncover pain points that users might not articulate in a survey, and build empathy for the user's daily reality.

2. Questionnaire/Survey

A 10-question online survey (see Appendix C) was distributed to a neurodivergent community group, yielding 8 responses. The survey used a mix of Likert-scale items and open-ended prompts to validate and quantify patterns observed in interviews. This broader sample confirmed that the problems identified were widespread, not limited to the two interview participants.

3. Document Analysis

The team reviewed published literature on sensory processing differences, occupational therapy dressing strategies, and the feature sets of competing apps (Stylebook, Cladwell, standard weather apps). This ensured the solution was informed by clinical best practices and clearly differentiated from existing tools.

Why these techniques were chosen: Interviews provided deep qualitative insight that reveals the "why" behind user behavior. The survey broadened the sample size to validate that patterns were not idiosyncratic to a few individuals. Document analysis grounded the team in the competitive landscape and clinical context, ensuring the product addressed real gaps rather than reinventing features that already exist elsewhere.

b. Data Collection Participants

- **Interview participants (2 total):** Three sensory-sensitive adults (both aged 31, diagnosed with autism or ADHD) recruited through the developers' personal circles.
- **Survey respondents (8 total):** Members of an online neurodivergent community who voluntarily completed the anonymous survey. This was a convenience sample chosen for direct relevance — all respondents self-identified as having sensory sensitivities or caring for someone who does.

- **Why these individuals were chosen:** The primary users of SensoryWardrobe are sensory-sensitive people and their caregivers. Gathering requirements directly from this population ensures the system addresses verified, real-world needs rather than developer assumptions. The occupational therapy clinic connection provided access to individuals actively managing sensory challenges, while the parent group offered caregiver perspectives essential for the multi-profile feature.

c. Supporting Documents

The following documents are attached as appendices to this requirements document:

- **Appendix A:** Interview Question Guide
- **Appendix B:** Interview Transcript Summaries (anonymized)
- **Appendix C:** Online Survey Questionnaire
- **Appendix D:** Survey Response Summary
- **Appendix E:** Competitive Feature Comparison Matrix

Appendix A: Interview Question Guide

1. Describe your typical morning routine for getting dressed. How long does it usually take?
2. What sensory factors most affect your clothing comfort (texture, pressure, weight, seams, tags, temperature)?
3. How do you currently decide what to wear? Do you consider the weather forecast?
4. Have you ever had a day ruined or significantly impacted by wearing the wrong clothing? Please describe.
5. Do you keep any record of what you wore and how comfortable it was? If so, how?
6. What information, if available at your fingertips each morning, would make getting dressed easier?
7. Have you tried any apps or tools to help with outfit planning? What did you like or dislike about them?
8. How would you feel about rating your outfit comfort at the end of each day on a 1–5 scale?
9. If an app could suggest outfits based on today's weather and your past comfort history, how useful would that be to you (1–5)?
10. Is there anything else about your clothing/sensory experience you think a helpful app should know?

Appendix B: Interview Transcript Summaries (Anonymized)

Participant 1 (Adult, age 31, autism diagnosis)

Reports spending very little time getting ready in the morning, but being significantly bothered by clothing choices if they don't match his sensory needs. Specifically reports issues with garment texture, seams, and tags. States that comfort is the sole factor in deciding what to wear, and has experienced days going off-track solely due to clothing-related issues. Currently uses no tracking system due to the effort required. Rated the concept of weather-linked outfit suggestions as moderately useful (3-4/5), indicating that he wouldn't use the weather portion, but the comfort tracking features would be very useful.

Participant 2 (Adult, age 31, ADHD diagnosis)

Experiences severe decision fatigue as a result of ADHD. Reports ability to get ready quickly due to lack of choices resulting from job requirements. Affected by pressure/tightness of clothing most of all. Reports prioritizing the fit of clothing over weather. Reports regular occurrences of difficult days caused exclusively by tight clothing but would not be particularly interested in an app for tracking.

Appendix C: Online Survey Questionnaire

1. Do you identify as sensory-sensitive, or do you care for someone who is? (Yes / No / Prefer not to say)
2. On a scale of 1–5, how much does clothing choice affect your daily mood and productivity? (1 = not at all, 5 = significantly)
3. How long does it typically take you to choose an outfit in the morning? (Less than 5 min / 5–10 min / 10–20 min / More than 20 min)
4. Which sensory factors most affect your clothing comfort? (Check all that apply: Texture/fabric feel, Pressure/tightness, Seams and tags, Weight/heaviness, Temperature sensitivity, Other — please specify)
5. Do you currently consider the weather forecast when choosing clothes? (Always / Sometimes / Rarely / Never)
6. Have you ever tracked your outfits and how comfortable they were? (Yes, on paper / Yes, digitally / No, never)
7. If you answered "No" to question 6, what prevented you from tracking? (Open-ended)
8. On a scale of 1–5, how useful would an app that suggests outfits based on today's weather and your past comfort history be to you?

9. How often would you be willing to spend 30–60 seconds rating your outfit's comfort? (Daily / A few times a week / Weekly / Rarely/Never)

10. Please rank the following features by importance to you (1 = most important, 6 = least important): Weather integration, Sensory tags on clothing, Comfort rating system, Smart outfit suggestions, Outfit history/calendar view, Accessibility options (dark mode, large buttons)

Appendix D: Survey Response Summary

Total respondents: 28

Question	Key Finding
Q1: Sensory-sensitive?	100% Yes (selection criteria)
Q2: Clothing impact on mood (1–5)	Mean: 3.875; 62.5% rated 4 or 5
Q3: Time choosing outfit	37.5% spend 5+ minutes; 25% spend 10+ minutes
Q4: Top sensory factors	Texture (75%), Seams/tags (62.5%), Pressure (62.5%), Weight (25%), Temperature (87.5%)
Q5: Consider weather?	50% Always, 50% Sometimes, 0% Rarely, 0% Never
Q6: Ever tracked outfits?	0% Yes (paper); 0% digitally; 100% Never
Q7: Why not track?	"Never thought to" (50%), "Mentally track" (25%), "Would forget" (12.5%), "Immediately dispose of uncomfortable garments" (12.5%)
Q8: App usefulness (1–5)	Mean: 3.125; 37.5% rated 4 or 5
Q9: Willing to rate daily?	37.5% Daily (if under 60 sec); 37.5% A few times/week
Q10: Feature priority ranking	1. Comfort rating, 2. Sensory tags, 3. Weather integration, 4. Accessibility, 5. History view, 6. Smart suggestions

Appendix E: Competitive Feature Comparison Matrix

Feature	Sensory Wardrobe	Paper Journal	Spreadsheet	Stylebook	Cladwell	Weather Apps
Automatic weather fetch	✓	✗	✗	✗	✓	✓
Sensory-specific tags	✓	Possible (manual)	Possible (manual)	✗	✗	✗
Comfort rating system	✓	Possible (manual)	Possible (manual)	✗	✗	✗
Weather-to-comfort correlation	✓	✗	Difficult/manual	✗	✗	✗
Smart outfit suggestions	✓	✗	✗	✗	Partial (style-based)	✗
Searchable history	✓	✗	✓	✓	✓	✗
Low daily effort (<60 sec)	✓	✗	✗	✓	✓	N/A
Accessibility-first design	✓	N/A	✗	✗	✗	Varies
Multi-profile (caregiver)	✓	✗	Possible	✗	✗	✗
Mobile-native experience	✓	✗	✗	✓	✓	✓
No ongoing cost to user	✓	✓	✓ (free tier)	✗ (\$30)	✗ (subscription)	✓
Works offline	✓	✓	✗	✓	✗	✗